

Azure Monitor Alert Action Group and On-Call Routing

Project Resource Group: rg-monitor-project **Target Virtual Machine:** monitor-vm
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ABSTRACT

In modern cloud environments, maintaining high availability and rapid response times to infrastructure issues is critical. This project implements a centralized monitoring and automated notification pipeline within Microsoft Azure. By utilizing Azure Monitor, a Log Analytics Workspace, and Data Collection Rules, the system continuously tracks the CPU utilization of a target virtual machine. When resource thresholds are breached, Azure Action Groups orchestrate on-call routing to deliver targeted notifications directly to the responsible team members.

OBJECTIVE

The primary objective of this project is to demonstrate how cloud infrastructure can automatically detect operational anomalies and streamline incident response. Specific goals include:

- Configuring automated performance tracking for an Azure Virtual Machine.
- Collecting and storing performance telemetry using a Log Analytics Workspace and Data Collection Rule.
- Setting up threshold-based alerts to detect high CPU usage exceeding 80%.
- Implementing Azure Action Groups with separate routing configurations to ensure notifications reach the correct teams efficiently.

PROJECT ARCHITECTURE & DETAILS

Resource Group	rg-monitor-project serves as the logical container holding all deployed Azure components for this project.
Virtual Machine	monitor-vm is the primary cloud workload configured to generate operational metrics.
Monitoring & Collection	Azure Monitor tracks CPU utilization, utilizing a Data Collection Rule to gather telemetry into a Log Analytics Workspace.
Alerting Mechanism	A CPU alert rule is actively configured to trigger whenever the CPU utilization of monitor-vm goes above 80%.
Notifications & Routing	Azure Action Groups receive alert triggers and execute separate routing pathways to notify designated teams or on-call personnel.

EXPECTED OUTCOME

The successful completion of this project demonstrates that Azure can automatically detect